Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Contrasts on the Exchange-Aligned Data Model

PENPAL Project

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Contents

This notebook compares **valence alignment** across the three experimental conditions using the new exchange-aligned structure.

- **Human-AI**: Human co-writing with an LLM
- **Human-Human**: Two humans co-writing together
- **AI-AI**: Two AI agents co-writing together

Important structural note:

- `author_1` / `author_2` are now treated as **slot 1 / slot 2** in the cleaned exchange sequence.
- In **Human-AI**, those slots also map onto human vs AI roles.
- In **Human-Human** and **AI-AI**, the slots are positional sides of the exchange, not stable identity labels.

The analysis window is restricted to **complete exchanges** and the default aligned turn range (`analysis_turn` 1 to 9).

Primary cross-condition object. Each story contributes one unsigned, label-invariant dyad asymmetry magnitude:

$$A_{\text{valence}}(i) \;=\; \big|\,\text{atanh}(r_{1\to 2,i}) - \text{atanh}(r_{2\to 1,i})\,\big|$$

where $r_{1\to 2,i}$ is the per-story Pearson correlation between same-turn slot 1 and slot 2 valence, and $r_{2\to 1,i}$ is the correlation between slot 2's previous-turn valence and slot 1's current-turn valence. The main question is whether A_{valence} differs across conditions. The directional-slope LMMs are retained as supplemental views. Starter-mechanism and Human-AI speaker-type decompositions are deferred to a follow-up pass.

1 Load Data

```
## Data availability:

## # A tibble: 6 x 4
##   file                                     `human-ai` `human-human` `ai-ai`
##   <chr>                                    <lgl>      <lgl>         <lgl>
## 1 dyadic_sentiment_scores.parquet          TRUE       TRUE          TRUE
## 2 interaction_level_surface_metrics.parquet TRUE      TRUE          TRUE
## 3 novelty_scores.csv                       TRUE       TRUE          TRUE
## 4 semantic_exploration_binned.parquet      TRUE       TRUE          TRUE
## 5 story_embeddings_interaction_level.parquet TRUE     TRUE          TRUE
## 6 interaction_level_stories_filtered.csv   TRUE       TRUE          TRUE
```

Table 1: Loaded exchange-aligned valence data by condition

condition	n_stories	n_turns	min_turn	max_turn
Human-AI	48	431	1	9
Human-Human	31	279	1	9
AI-AI	39	351	1	9

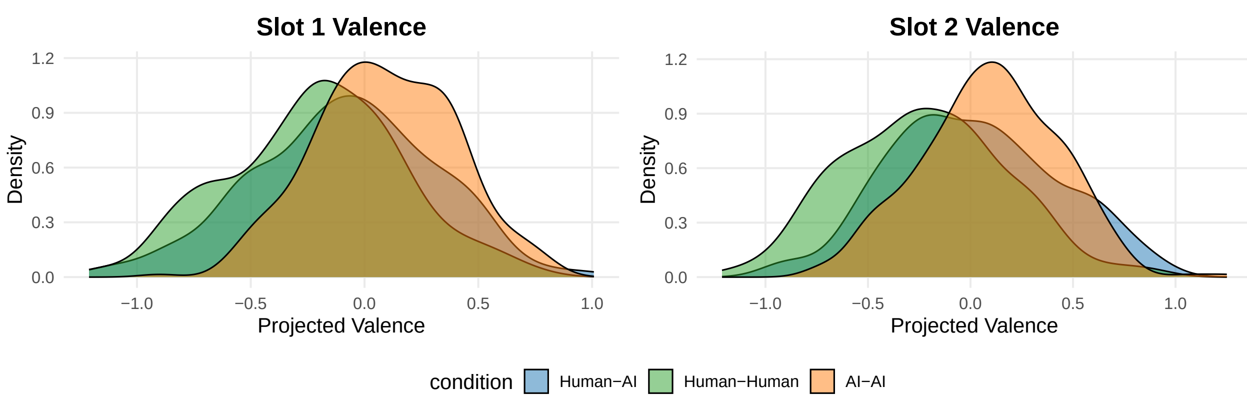
2 Preprocessing

```
## Analysis rows: 1886

## Stories: 118
```

3 Descriptive Checks

Table 2: Mean projected valence by condition

condition	n_stories	n_turns	slot_1_mean	slot_2_mean
Human-AI	48	383	-0.088	0.013
Human-Human	31	248	-0.214	-0.223
AI-AI	39	312	0.089	0.096



4 Primary: Per-Story Asymmetry Magnitude

```
## Stories with both directional correlations available:
```

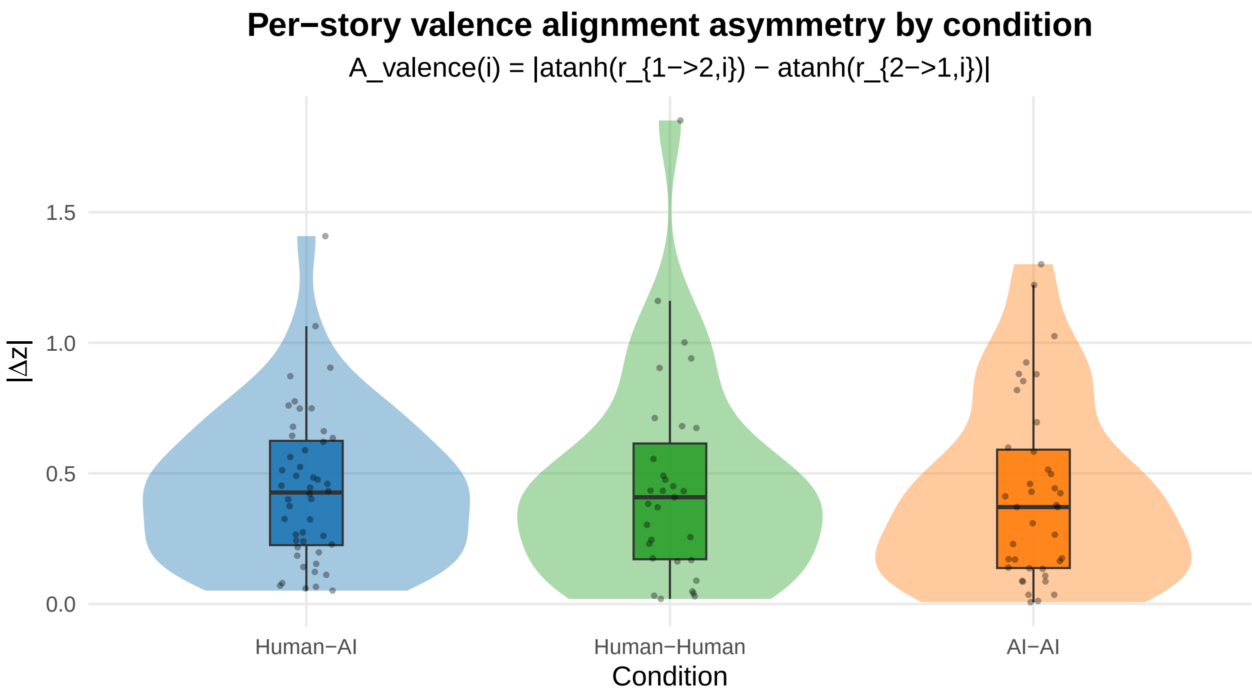
Table 3: Per-story A_valence sample sizes by condition

condition	n
Human-AI	48
Human-Human	31
AI-AI	39

Table 4: Per-story |Fisher-z asymmetry| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	48	0.441	0.289	0.427
Human-Human	31	0.457	0.400	0.408
AI-AI	39	0.421	0.348	0.371

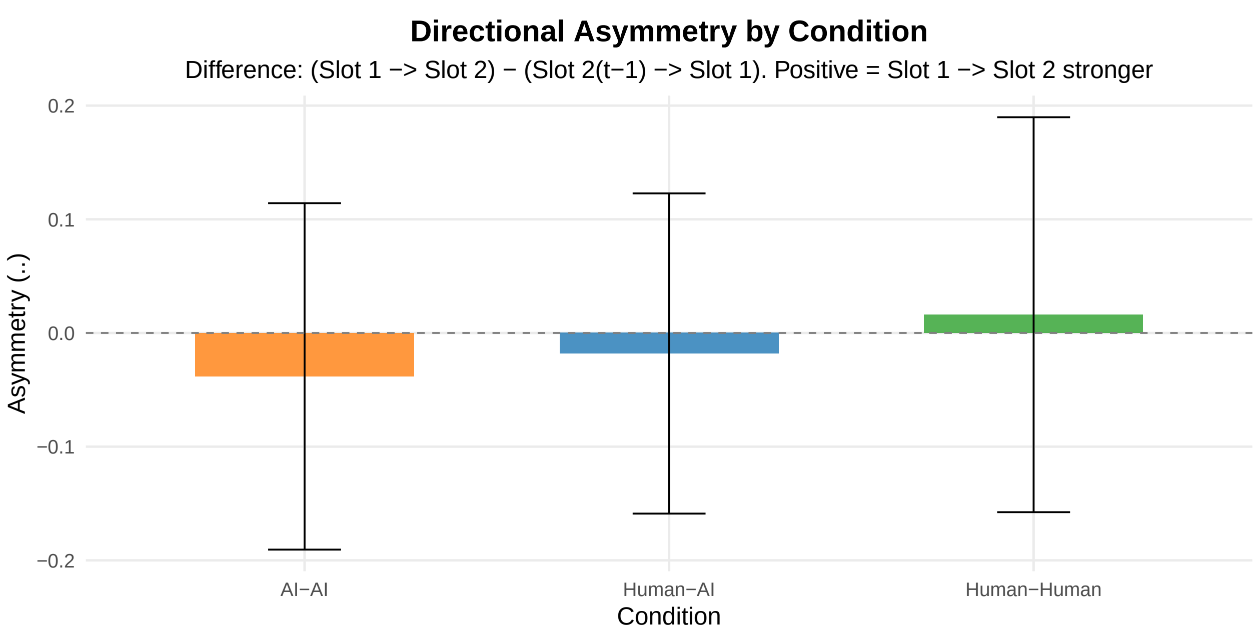
```
## One-way ANOVA: A_valence ~ condition

##              Df Sum Sq Mean Sq F value Pr(>F)
## condition     2  0.022 0.01101   0.095  0.909
## Residuals   115 13.313 0.11576

##
## Kruskal-Wallis (robust): A_valence ~ condition

##
##  Kruskal-Wallis rank sum test
##
## data:  A_valence by condition
## Kruskal-Wallis chi-squared = 0.65297, df = 2, p-value = 0.7215

##
## Pairwise Wilcoxon (BH-adjusted):

##
##  Pairwise comparisons using Wilcoxon rank sum exact test
##
## data:  df_asym$A_valence and df_asym$condition
##
##             Human-AI Human-Human
## Human-Human 0.73     -
## AI-AI       0.73     0.73
##
## P value adjustment method: BH
```



5 Supplemental: Within-Condition Directional Asymmetry

Table 5: Within-condition alignment asymmetry by slot direction

condition	slope_slot1_to_slot2	slope_slot2_to_slot1	difference	se	t	p	sig
Human-Human	0.143	0.159	0.016	0.089	0.181	0.856	
Human-AI	0.135	0.117	-0.018	0.072	-0.251	0.802	
AI-AI	0.193	0.154	-0.038	0.078	-0.491	0.623	



6 Supplemental: Cross-Condition Directional Model

```
## Full model summary:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: response_z ~ predictor_z * alignment_direction * condition +
##     (1 + predictor_z | conversation_id)
##    Data: df_std
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 5119.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.55075 -0.74244 -0.02974  0.72945  2.54414
##
## Random effects:
##  Groups          Name        Variance Std.Dev. Corr
##  conversation_id (Intercept) 0.00000  0.0000
##                  predictor_z 0.01426  0.1194    NaN
##  Residual                    0.84830  0.9210
## Number of obs: 1886, groups:  conversation_id, 118
##
## Fixed effects:
##                                          Estimate Std. Error        df
## (Intercept)                             1.591e-18  2.154e-02  1.759e+03
## predictor_z                             1.504e-01  2.560e-02  1.151e+02
## alignment_direction1                   -2.798e-18  2.154e-02  1.759e+03
## condition1                              3.869e-18  2.887e-02  1.759e+03
## condition2                             -1.349e-17  3.216e-02  1.759e+03
## predictor_z:alignment_direction1        6.689e-03  2.303e-02  1.759e+03
## predictor_z:condition1                 -2.404e-02  3.430e-02  1.152e+02
## predictor_z:condition2                  9.182e-04  3.821e-02  1.150e+02
## alignment_direction1:condition1         2.948e-18  2.887e-02  1.759e+03
## alignment_direction1:condition2         1.426e-17  3.216e-02  1.759e+03
## predictor_z:alignment_direction1:condition1  2.326e-03  3.086e-02  1.759e+03
## predictor_z:alignment_direction1:condition2 -1.473e-02  3.438e-02  1.759e+03
##                                         t value Pr(>|t|)
## (Intercept)                               0.000    1.000
## predictor_z                               5.877 4.15e-08 ***
## alignment_direction1                      0.000    1.000
## condition1                                0.000    1.000
## condition2                                0.000    1.000
## predictor_z:alignment_direction1          0.290    0.772
## predictor_z:condition1                   -0.701    0.485
## predictor_z:condition2                    0.024    0.981
## alignment_direction1:condition1           0.000    1.000
## alignment_direction1:condition2           0.000    1.000
## predictor_z:alignment_direction1:condition1   0.075    0.940
## predictor_z:alignment_direction1:condition2  -0.428    0.668
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##             (Intr) prdct_ algn_1 cndtn1 cndtn2 pr_:_1 prd_:1 prd_:2 al_1:1
## predictor_z  0.000
## algnmnt_dr1  0.000  0.000
## condition1  -0.153  0.000  0.000
## condition2   0.153  0.000  0.000 -0.512
## prdctr_z:_1  0.000  0.000  0.000  0.000  0.000
## prdctr_z:c1  0.000 -0.153  0.000  0.000  0.000  0.000
## prdctr_z:c2  0.000  0.153  0.000  0.000  0.000  0.000 -0.512
## algnmnt_1:1  0.000  0.000 -0.153  0.000  0.000  0.000  0.000  0.000
## algnmnt_1:2  0.000  0.000  0.153  0.000  0.000  0.000  0.000  0.000 -0.512
## prdct_:_1:1  0.000  0.000  0.000  0.000  0.000 -0.153  0.000  0.000  0.000
## prdct_:_1:2  0.000  0.000  0.000  0.000  0.000  0.153  0.000  0.000  0.000
##             al_1:2 p_:_1:1
## predictor_z
## algnmnt_dr1
## condition1
## condition2
## prdctr_z:_1
## prdctr_z:c1
## prdctr_z:c2
## algnmnt_1:1
## algnmnt_1:2
## prdct_:_1:1  0.000
## prdct_:_1:2  0.000 -0.512
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
##
## Alignment slopes by condition and direction:
```

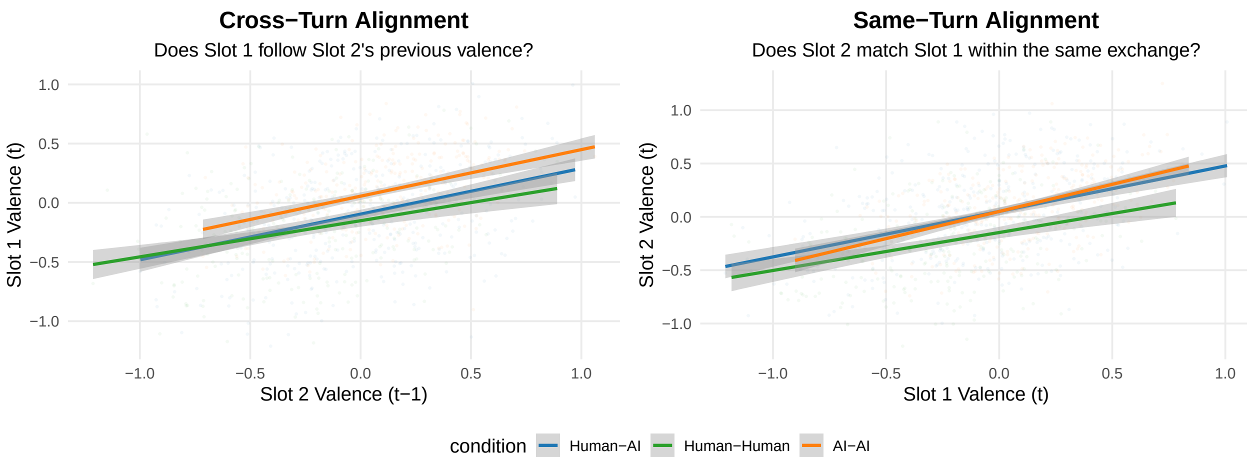
```
##  condition   alignment_direction  predictor_z.trend     SE  df lower.CL
##  Human-AI    Slot 1 → Slot 2                  0.135 0.0532 361   0.0308
##  Human-Human Slot 1 → Slot 2                  0.143 0.0661 359   0.0133
##  AI-AI       Slot 1 → Slot 2                  0.193 0.0589 359   0.0768
##  Human-AI    Slot 2(t-1) → Slot 1             0.117 0.0532 361   0.0128
##  Human-Human Slot 2(t-1) → Slot 1             0.159 0.0661 359   0.0294
##  AI-AI       Slot 2(t-1) → Slot 1             0.154 0.0589 359   0.0386
##  upper.CL
##     0.240
##     0.273
##     0.309
##     0.222
##     0.289
##     0.270
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

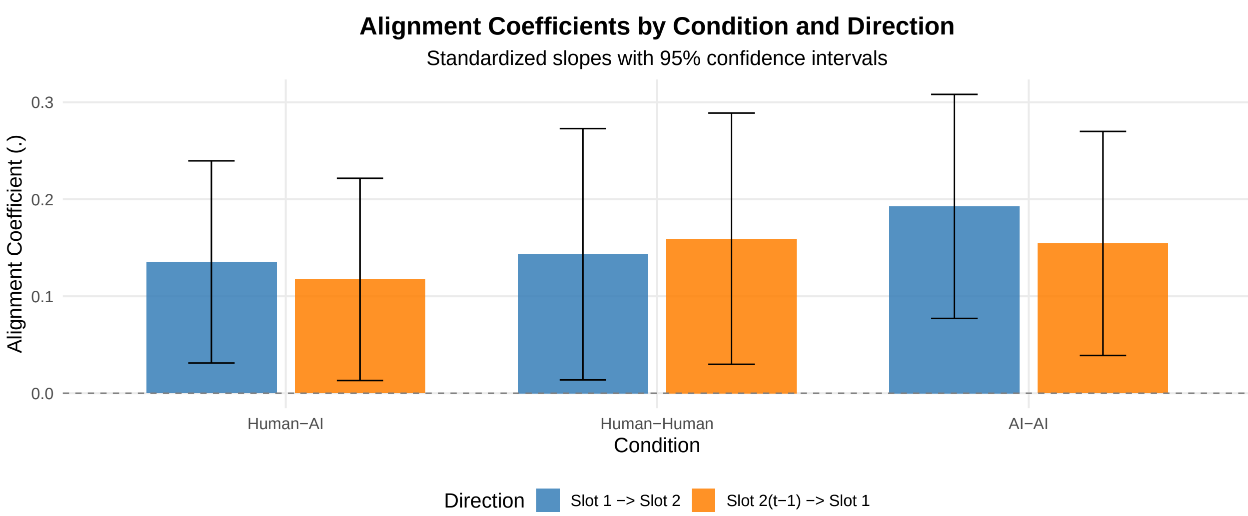
```
##
## Within-condition direction contrasts:
```

```
## condition = Human-AI:
##  contrast                                  estimate     SE   df t.ratio p.value
##  (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2  -0.0180 0.0712 1644  -0.253  0.8000
##
## condition = Human-Human:
```

```
##   contrast                                    estimate     SE   df t.ratio p.value
##   (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2    0.0161 0.0884 1644   0.182  0.8557
##
## condition = AI-AI:
##   contrast                                    estimate     SE   df t.ratio p.value
##   (Slot 2(t-1) → Slot 1) - Slot 1 → Slot 2   -0.0382 0.0788 1644  -0.484  0.6282
##
## Degrees-of-freedom method: kenward-roger
```

7 Supplemental: Directional Scatter Visualizations





8 Summary

Primary cross-condition result: per-story $A_{\text{valence}} = |\operatorname{atanh}(r_{1\to 2,i}) - \operatorname{atanh}(r_{2\to 1,i})|$, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Supplemental views retained:

- within-condition directional slope LMMs (two pairing rules)
- full cross-condition LMM with direction 00d7 condition interactions
- directional scatterplots

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)
- Human-AI decomposition (`speaker_type \u00d7 speaker_is_starter` within Human-AI only)

Analysis completed: 2026-04-23 12:20:12.833962